3. $\frac{1}{4}; \frac{1}{2}$ 4. (a) $\frac{1}{9}$ (b) $\frac{7}{9}$ 5. (i) $\frac{3}{5}, \frac{6}{5}$ (ii) $\frac{2}{25}$ 7. 1 9. $\frac{3}{4}, \frac{27}{80}$ 10. $\frac{1}{2}$

7. Probability distributions

Exercise: 7.1

6. (a) 0.059 (b) 0.2642 (c) 0.0133 (d) mean = 1 and variance = 0.95

7. (i) 0.01008 (ii) 0.000262 (iii) 0.09935 8. 0.375 9. 0.65536

10. (i) 0.3969 (ii) 0.45212 (iii) 0.9797 11. 5 or more trials 12. 0.7530

13. (i) 703 (ii) 516 (iii) 656 14. (i) 0.0634 (ii) 0.0634 (iii) 0.9729

15. 25/216 16. $\binom{25}{x}\left(\frac{1}{5}\right)^{x}\left(\frac{4}{5}\right)^{(25-x)}$ 17. $\frac{3}{4^{14}}$ 18. 0.2626 19. 0.8743

20. (i) $\frac{80}{243}$ (ii) $\frac{192}{243}$

Exercise: 7.2

6. 0.2352 7. 0.0025 8. (i) 0.2231 (ii) 0.1912

9.(i) 0.08208 (ii) 0.2138 (iii) 0.1089 10. (i) 2 days (ii) 91 days (iiii) 43 days

11. 0.0265 12. (i) 0.1353 (ii) 0.3235

Exercise: 7.3

5. (i) 67 (ii) 134 (iii) 1637 6 (i) mean = 60.48 (ii) standard deviation = 19.78

7. (i) 0.9772 (ii) 0.49865 8. (a) 46 (b) 46 (c) 342

9. 0.719 10. (i) 0.2420 (ii) 0.8413

Exercise 7.4

1	2	3	4	5	6	7	8	9	10	11	12	13	14
(b)	(c)	(c)	(c)	(c)	(a)	(b)	(a)	(c)	(d)	(a)	(a)	(d)	(b)
15	16	17	18	19	20	21	22	23	24	25	26	27	28
(d)	(b)	(d)	(d)	(d)	(d)	(a)	(a)	(b)	(b)	(a)	(c)	(d)	(d)

Miscellaneous problems

1. (i) 0.89131 (ii) 0.34173 2. 0.03295 3. 0.98981

4. 0.0067379 or 6.7379×10^{-3} 5. 80.33%

6. a) 0.4013 (b) 0.3413 7. a) 30.85% b) 37.20% c) 10.56%

8. a) 0.9938 (b) 0.9878 (c) 0.3944 9. 0.2119 10. 7

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